# 4

## Entangling Space

### WHAT DOES IT MEAN TO BE SEPARATE IN A QUANTUM UNIVERSE?

To accept special and general relativity is to abandon Newtonian absolute space and absolute time. While it's not easy, you can train your mind to do this. Whenever you move around, imagine your now shifting away from the nows experienced by all others not moving with you. While you are driving along a highway, imagine your watch ticking away at a different rate compared with timepieces in the homes you are speeding past. While you are gazing out from a mountaintop, imagine that because of the warping of spacetime, time passes more quickly for you than for those subject to stronger gravity on the ground far below. I say "imagine" because in ordinary circumstances such as these, the effects of relativity are so tiny that they go completely unnoticed. Everyday experience thus fails to reveal how the universe really works, and that's why a hundred years after Einstein, almost no one, not even professional physicists, feels relativity in their bones. This isn't surprising; one is hard pressed to find the survival advantage offered by a solid grasp of relativity. Newton's flawed conceptions of absolute space and absolute time work wonderfully well at the slow speeds and moderate gravity we encounter in daily life, so our senses are under no evolutionary pressure to develop relativistic acumen. Deep awareness and true understanding therefore require that we diligently use our intellect to fill in the gaps left by our senses.

While relativity represented a monumental break with traditional ideas about the universe, between 1900 and 1930 another revolution was also turning physics upside down. It started at the turn of the twentieth century with a couple of papers on properties of radiation, one by Max Planck and the other by Einstein; these, after three decades of intense research, led to the formulation of *quantum mechanics.* As with